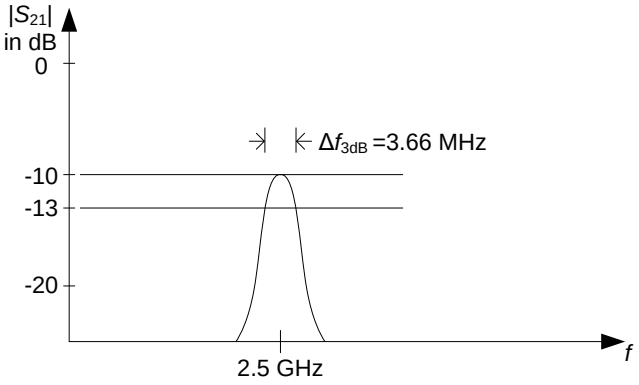
Winter term 2020/2021: Exercises "Radio Frequency Technology"

EXERCISE SESSION 12 (05.02.2021)

Problem 12.1 Phase noise_1

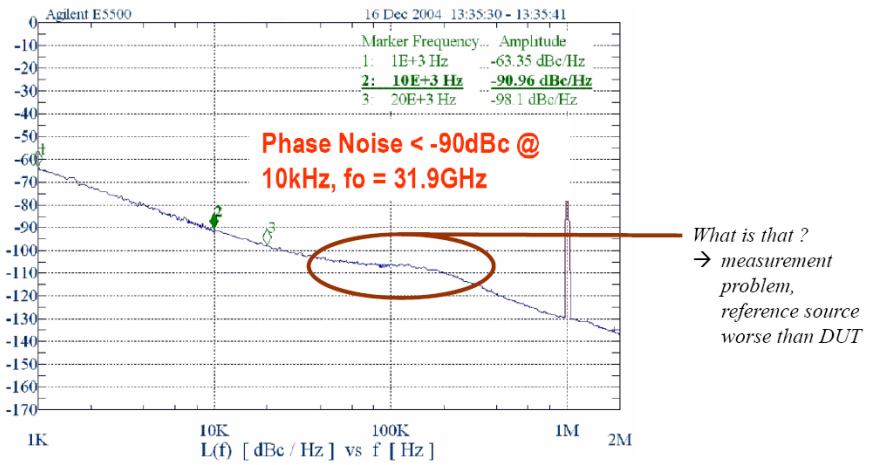
At 2.5 GHz microwave oscillator uses a loosely coupled λ/2-transmission line resonator for frequency stabilization. The resonator including its coupling to the rest of the oscillator circuit is measured separately, giving the $|S_{21}|$ plot as shown below.

a) Indicate the unloaded quality factor of the resonator.
b) In order to improve phase noise at offset frequency around 100 kHz, a cavity with known unloaded Q of 8000 is considered. It will be coupled to the oscillator circuit to the same level as the existing λ/2-resonator. Determine the improvement in phase noise, L(f_m), in dB.



Problem 12.2: Phase noise_2

For the millimeter-wave oscillator measured as shown below, the measured curve is inaccurate for offset frequencies beyond 20 kHz, since the reference source noise becomes worse than the DUT noise.



a) Assuming a -20 dB/decade slope for offsets from 20 kHz to 100 kHz, calculate the to-be-expected phase noise at 100 kHz offset.
b) Considering a resonator Q of 2800 (known from measurements of the separate resonator) and a noise figure of the amplifier of 3 dB (datasheet value), calculate the signal power at the input of the amplifier (in dBm) at room temperature.

Winter term 2020/2021: Exercises "Radio Frequency Technology"

Problem 12.3: Network analyzer accuracy

A network analyzer operating at lower microwave frequency was calibrated using high-quality cal standards "open", "short" and "match". There are, however, statistical error sources which cannot be accounted for by calibration. Due to these errors, accuracy is not perfect. Verification measurements (measurements of known loads) showed, that a non-perfect load can be measured and the measurement of $|S_{11}| = -34$ dB be trusted. Loads with $|S_{11}|$ better than -34 dB cannot be measured with confidence.

a) Determine the impedance of a load resulting in $|S_{11}| = -34$ dB.
b) Using the same error margin as for the matched load, calculate the impedances of an "almost short" and an "almost open".

Short questions session 12

a) Which intermodulation signals (order of the intermodulation product) are more difficult to avoid in a communications system and why is this so?

b) You have to detect very small signals which are near the noise floor of your spectrum analyzer. By which means can you increase the readability (SNR) of such signals?